$A \in \mathbb{C}^{m\times n}$ $\quad \alpha, \beta \in \mathbb{C}$ $\quad \vec{x}, \vec{y} \in \mathbb{C}^n$

$$A(\alpha\vec{x} + \beta\vec{y}) = \alpha A\vec{x} + \beta A\vec{y}$$

The action of multiplying A on a column vector in $\mathbb{C}^n$ is "linear".

We consider a mapping from $\mathbb{C}^n$ to $\mathbb{C}^m$ such that

$$\begin{array}{ccc} \vec{x} & \longmapsto & A\vec{x} \\ \cap & & \cap \\ \mathbb{C}^n & & \mathbb{C}^m \end{array}$$

linear operator

$A \in \mathbb{C}^{m\times n}$ $\qquad A^H$: Hermitian transpose

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \qquad A^H = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$\vec{x} \in \mathbb{C}^n$ $\quad \vec{y} \in \mathbb{C}^m$ $\qquad \vec{y} \cdot A\vec{x} = A^H\vec{y} \cdot \vec{x}$

$$\vec{y} \cdot A\vec{x} = \vec{y}^H A\vec{x}$$

$$A^H\vec{y} \cdot \vec{x} = (A^H\vec{y})^H \vec{x} = \vec{y}^H A\vec{x}$$

X, Y : vector spaces with inner product.

Let $A : X \longrightarrow Y$ be a linear operator.

A^* : adjoint of A satisfies

$$y \cdot Ax = A^* y \cdot x \qquad \forall x \in X, \ \forall y \in Y$$

$X = Y$ = vector space of sequences
OR
DT signals.

$x[n], y[n] \in X$

$$x[n] \cdot y[n] = \sum_{k=-\infty}^{\infty} x^*[k]\, y[k]$$

D : delay operator

$$x[n] \longmapsto D\{x[n]\} = x[n-1] \qquad \text{linear operator.}$$

$$\begin{bmatrix} \ddots & \ddots & & & & \\ & 0 & 1 & & & \\ & & 0 & 1 & & \\ \cdots & & & 0 & 1 & \cdots \\ & & & & 0 & 1 \\ & & & & & 0 \quad 1 \\ & & & & & \ddots \end{bmatrix}$$

$k-1 = m$

$$y[n] \cdot D\{x[n]\} = y[n] \cdot x[n-1] = \sum_k y^*[k]\, x[k-1] = \sum_m y^*[m+1]\, x[m]$$

$$\underbrace{D^*\{y[n]\}}_{z[n]} \cdot x[n] = \sum_k z^*[k]\, x[k]$$

$$z[n] = y[n+1]$$

$$D^*\{y[n]\} = y[n+1]$$

A, B matrices of compatible sizes.

$$(AB)^H = B^H A^H$$

A, B : linear operators compatible.

$$(AB)^* = B^* A^*$$

$$D^3 = D \cdot D \cdot D$$

$$(D^3)^H = D^H D^H D^H = (D^H)^3$$

$$\mathcal{H} : x[n] \longmapsto \sum_k h[k]\, x[n-k]$$ convolution operator.

$$y[n] \cdot \mathcal{H}\{x[n]\} = y[n] \cdot \sum_k h[k]\, x[n-k]$$

$$= \sum_n y^*[n] \sum_k h[k]\, x[n-k]$$

$$= \sum_n y^*[n] \sum_k x[k]\, h[n-k]$$

$$= \sum_k \left(\sum_n y^*[n]\, h[n-k] \right) x[k]$$

$$= \sum_k \underbrace{\left(\sum_n y[n]\, h^*[n-k] \right)}_{z[k]}^* x[k]$$

$$z[k] = \sum_n y[n]\, h^*[n-k]$$

$$\boxed{\tilde{h}[n] = h^*[-n]}$$

conjugate filter.

$$= y[k] * \tilde{h}[k]$$

$$\mathcal{H}^* \{y[n]\} = \tilde{h}[n] * x[n]$$

$y_1[n] = h_1[n] * x[n]$

$y_2[n] = h_2[n] * x[n]$

$h_1[n], h_2[n]$: FIR of length L_h

$x[n]$: FIR length L_x

$y_1[n], y_2[n]$: FIR length $L_y = L_x + L_h - 1$

$$h_1[n] * y_2[n] = y_1[n] * h_2[n]$$

$$T_{y_2}\{h_1[n]\} = T_{y_1}\{h_2[n]\}$$

$$\begin{bmatrix} T_{y_2} & -T_{y_1} \end{bmatrix} \begin{pmatrix} h_1[n] \\ h_2[n] \end{pmatrix} = 0$$

$$\underbrace{\left(\begin{bmatrix} T_{y_2}^* \\ -T_{y_1}^* \end{bmatrix} \begin{bmatrix} T_{y_2} & -T_{y_1} \end{bmatrix} \right)}_{A}$$

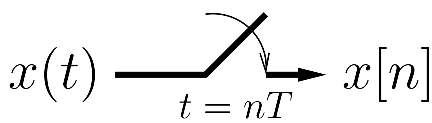
Sampling and Reconstruction:

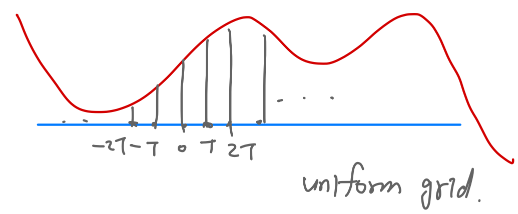
- Until now, we have considered continuous-time signals & systems *separately* from discrete-time signals & systems.
- We will now connect them through *uniform sampling*:

$$x[n] = x(nT) \text{ for } n \in \mathbb{Z},$$

Sampling period

where T (in seconds/sample) is the *sampling interval* and so $1/T$ is the *sampling rate*. Hz





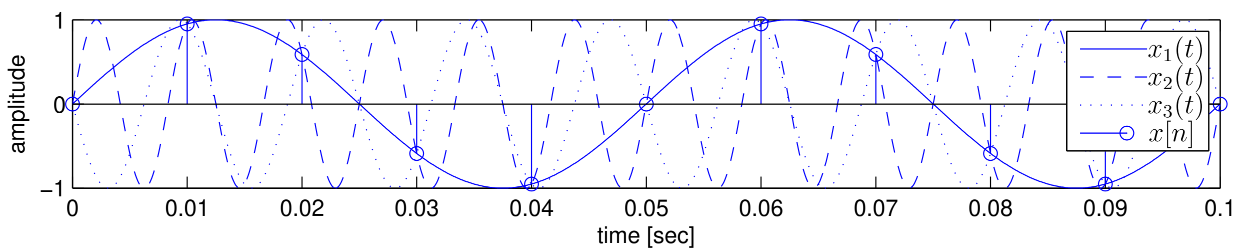
- Key question:
  - Do we always lose information in going from $\{x(t)\}_{\forall t}$ to $\{x[n]\}_{n=-\infty}^{\infty}$?
  - Under what conditions can we reconstruct $\{x(t)\}_{\forall t}$ from $\{x[n]\}_{\forall n}$?

Aliasing — intuitions:

- Consider sampling the 3 sinusoids

$$\begin{aligned} x_1(t) &= \sin(2\pi 20t) \\ x_2(t) &= \sin(2\pi 120t) \\ x_3(t) &= \sin(2\pi(-80)t) \end{aligned}$$

at a rate of $1/T = 100$ samples/sec to give $x[n]$:



They all yield the same $x[n]$!

- More generally, consider sampling a complex sinusoid:

$$x(t) = e^{j\Omega_k t} \quad \xrightarrow{t=nT} \quad x[n] = e^{j\Omega_k Tn}.$$

Which different Ω_k yield identical sequences $\{x[n]\}$?

Consider

$$\Omega_k \triangleq \Omega_0 + \frac{2\pi k}{T} \text{ for any integer } k.$$

$e^{j2\pi} = 1 \qquad e^{j2\pi kn} = 1$

Then

$$e^{j\Omega_k Tn} = e^{j(\Omega_0 Tn + 2\pi kn)} = e^{j\Omega_0 Tn} = e^{j\omega_0 n},$$

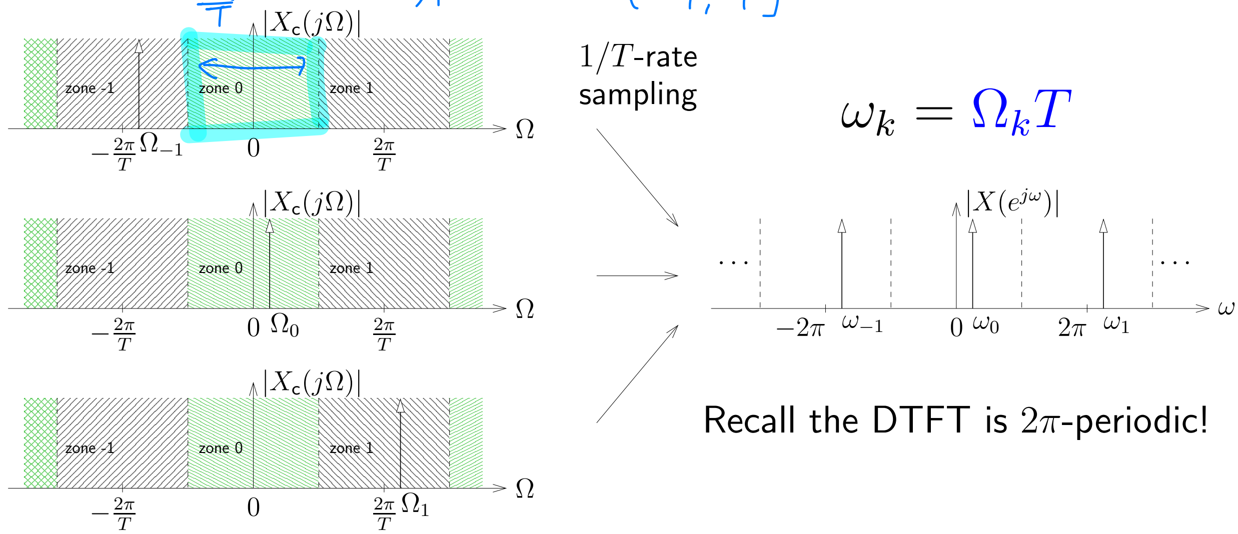
where $\boxed{\omega_0 = \Omega_0 T,}$ for any value of of k.

Aliasing — Nyquist zones:

- So far, we have seen that a sampled sinusoid $x[n]$ can be associated with *many* continuous-time sinusoids $x(t)$.
- Can visualize this for $x(t) = e^{j\Omega_k t}$ via "Nyquist zones":

  Defn: The k^{th} Nyquist zone is $\Omega \in \left(\frac{2\pi k - \pi}{T}, \frac{2\pi k + \pi}{T}\right]$

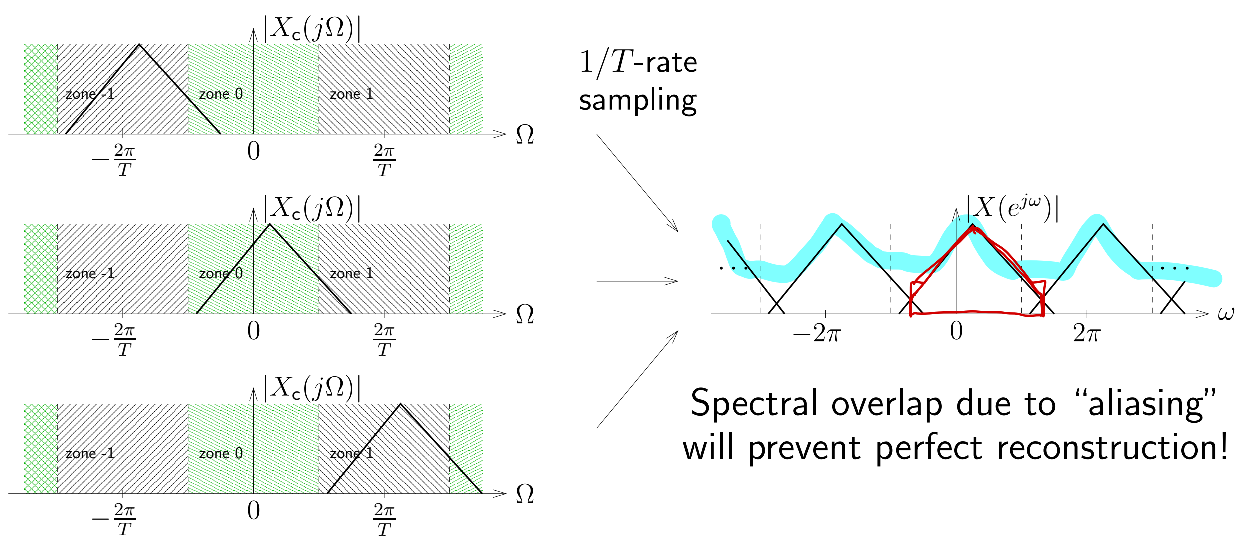
  $\frac{2\pi}{T}$ 0-th Nyquist zone $\left(-\frac{\pi}{T}, \frac{\pi}{T}\right]$



$$\omega_k = \Omega_k T$$

Recall the DTFT is 2π-periodic!

- Due to the linearity of sampling, the same phenomenon applies to generic signals:



Spectral overlap due to "aliasing" will prevent perfect reconstruction!

- Nyquist zone locations determined by sampling rate $\frac{1}{T}$!

$$T\{x(t)\} = x[n] = x(t)\Big|_{t=Tn}$$

$$T\{ax(t) + by(t)\} = a\,x[n] + b\,y[n]$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\Omega)\,e^{j\Omega t}\,d\Omega$$

Nyquist sampling:

- As we have seen, sampling can introduce ambiguity:

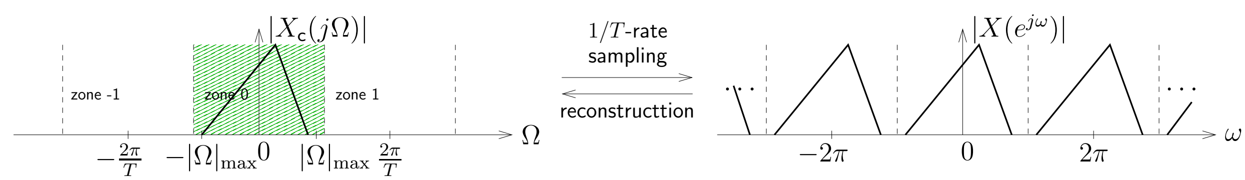
  Can't distinguish one Nyquist zone from another!

- How can we avoid this ambiguity?

  Force the CT signal to live in a single Nyquist zone via careful choice of sampling frequency $1/T$.

  There are two basic ways to do this:

  1. *Nyquist sampling*:
     - Ensure that all frequency components of $x(t)$ are in the 0^{th} Nyquist zone. Equivalently...
     - Sample more than twice as fast as the largest absolute signal frequency: $\frac{1}{T} > \frac{|\Omega|_{\max}}{\pi} = 2|f|_{\max}$



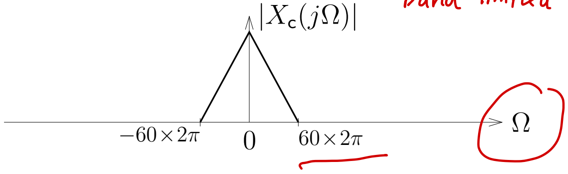
  2. *Bandpass sampling*:
     - Ensure than all frequency components of $x(t)$ are contained within some Nyquist zone k, for $k \neq 0$.
     - Note: works only for bandpass $x(t)$.
     - But how does one choose $\frac{1}{T}$ and k in this case?

- Reconstruction then boils down to optimal interpolation. Stay tuned for details...

Nyquist sampling – example:

- Suppose that $X_c(j\Omega)$ has the CTFT magnitude sketched below, where $|f|_{\max} = 60$ Hz.
- Nyquist sampling specifies that $\frac{1}{T} > 2 \times 60 = 120$ Hz.
- The figures below illustrate the Nyquist zones and the resulting DTFT for several choices of sampling rate $\frac{1}{T}$.

band limited signal.

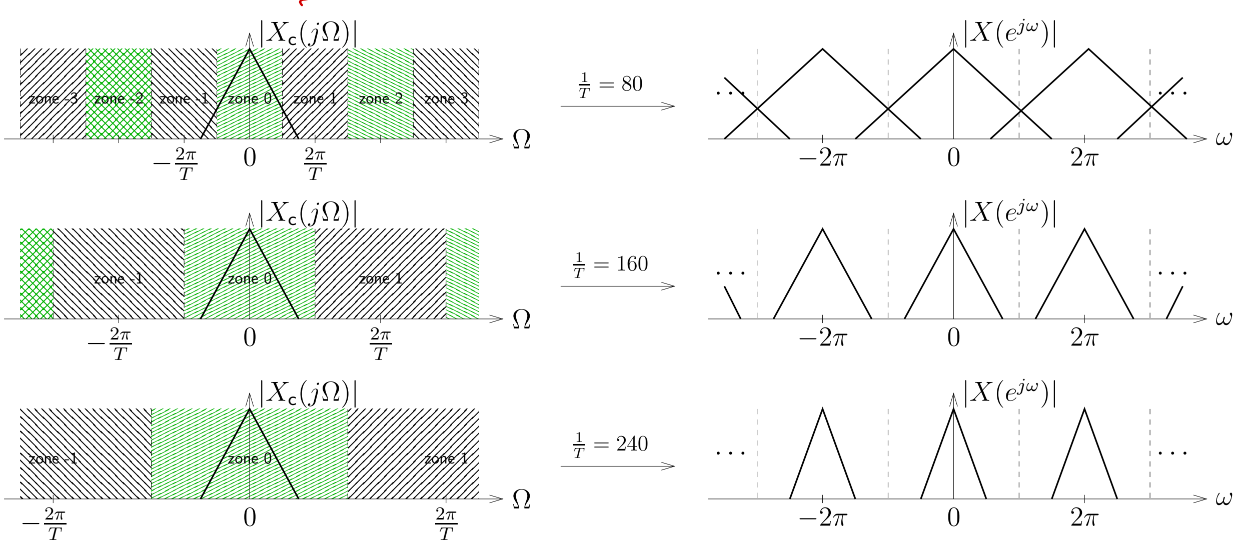


$\omega_{max} \le \frac{\pi}{T}$

$f_{max} \le \frac{1}{2T}$

$\omega_{max} = 2\pi f_{max}$

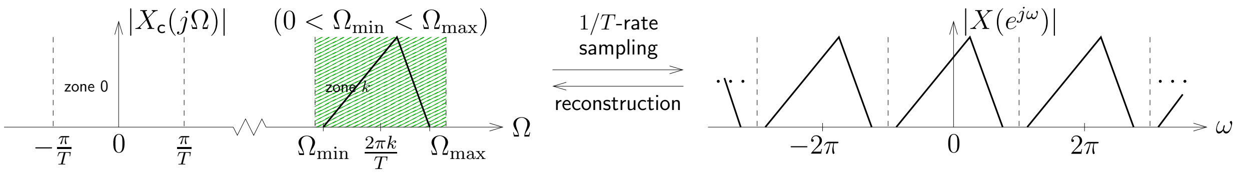
rad/sec Hz



- Note the presence of aliasing in the DTFT when $\frac{1}{T}$ is less than the Nyquist sampling rate (i.e., $2|f|_{\max}$ Hz).

Bandpass sampling – complex-valued signals:

- Consider a complex-valued bandpass signal of the form



- For practical reasons, we want to find the *smallest* possible sampling freq $\frac{1}{T}$. Then how do we choose k?

- First, note that fitting into the k^{th} Nyquist zone requires

capture the spectrum within the kth Nyquist zone.

$$\left\{\begin{array}{l}\frac{2\pi k-\pi}{T} < \Omega_{\min} \\ \Omega_{\max} \le \frac{2\pi k+\pi}{T}\end{array}\right\} \overset{\Omega=2\pi f}{\iff} \frac{k-0.5}{f_{\min}} < T \le \frac{k+0.5}{f_{\max}}.$$

- Also, note that T increases ($\frac{1}{T}$ decreases) as k increases. Thus, we want to find the largest integer k that satisfies

$$\frac{k-0.5}{f_{\min}} < \frac{k+0.5}{f_{\max}},$$

feasibility

and then use this "$k_{\max}$" to set the sampling interval as

$$T = \frac{k_{\max}+0.5}{f_{\max}}.$$

$f_{max} k - 0.5 f_{max} < f_{min} k + 0.5 f_{min}$

$k \le \left\lceil \frac{f_{min} + f_{max}}{2(f_{max} - f_{min})} \right\rceil - 1$

- After a bit of algebra, one can find

$$\frac{1}{T}\Big|_{\min} = \frac{f_{\max}}{\left\lceil \frac{1}{2}\frac{f_{\max}+f_{\min}}{f_{\max}-f_{\min}} \right\rceil - \frac{1}{2}}$$

where $\lceil\cdot\rceil$ denotes the "ceiling" operation, i.e., rounding up to the nearest integer.